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HornNet: A Two-Tier Deep Learning Framework for Real-Time Vehicle Classification from Acoustic Horn Signatures in Urban Environments

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***Abstract*—The study presents HornNet, a two-tier deep learning framework for real-time vehicle classification based on acoustic horn signatures in urban environments. The proposed work addresses the limitations of vision-based traffic monitoring systems. By leveraging hierarchical, Tier 1 performs lightweight horn detection using MFCC features, and Tier 2 executes detailed vehicle classification on Mel-spectrogram representations. HornNet achieved 96.2% horn detection accuracy, 94.7% vehicle classification accuracy, and 91.3% overall performance while reducing average Tier-2 inference operations by 78.3% through conditional activation. The model was evaluated on a dataset of 2,472 real-world horn recordings across six vehicle categories. THe HornNet outperforms conventional and deep learning baselines such as SVM, 1D-CNN, ResNet-50, and Transformer-CNN in both accuracy and inference speed. The model also maintained over 89% accuracy under heavy traffic and construction noise which confirms its robustness in complex acoustic conditions. Deployment tests on embedded devices (e.g, Raspberry PI, Coral TPU) validates its suitability for intelligent transportation and urban monitoring systems. It achieved inference times as low as 9.4 ms with low and stable power consumption (3.9–7.8 W depending on platform). Overall, HornNet establishes an efficient, scalable, and noise-resilient solution for real-time acoustic vehicle recognition in next-generation smart cities.**

***Index Terms*—Acoustic vehicle classification, deep learning, horn sound recognition, intelligent transportation systems (ITS), urban noise analysis, real-time embedded systems, edge computing.**

I. INTRODUCTION

Traffic congestion in cities have become a persistent problem in developing cities that experience high growth like Bangladesh. Serious pollution of noise caused by excessive movement of vehicles and uncontrolled usage of horns has caused low transportation efficiency as well as lowered transportation efficiency in the vehicles, which utilize them to produce honks [1], [2]. Ambient sound levels are often higher than the world health organization (WHO) range of 85 dB in megacities such as Dhaka with horn peaks go beyond 110 dB [3], [4]. Excessive horn usage contributes to noise pollution but also offers a distinct acoustic cue for traffic activity. Thus, horn-based sensing provides a viable alternative to vision-based ITS methods limited by occlusion, poor lighting, and high deployment cost.

Recent studies have advanced horn and vehicle detection using acoustic methods. Vehicle have been detected from sound reflections beyond blind corners, validating sound as a supplementary sensing modality under low-visibility conditions [5]. Prior studies have demonstrated strong performance in horn and siren detection using CNN [6]–[9], 1D-CNN [10], [11], and hybrid transformer architectures [12]–[14], achieving 90–99% accuracy across varied datasets. However, most works focus on binary detection tasks or emergency vehicle recognition rather than fine-grained multi-class vehicle classification in dense urban environments. Enhanced broader sound-event detection setups and benchmark datasets [15] [16], along with synthetic augmentation and feature-fusion techniques [17] further enhance robust performance in low signal-to-noise ratio environments. Collectively, the works demonstrate that CNNs, 1D-CNNs, and mixed acoustic models yield 90-99% accuracy with disparate datasets and motivate more works on horn-driven detection with high population density in urban landscapes.

Despite significant progress, key limitations persist in existing studies. Most prior works focus solely on binary horn or siren detection which overlooks fine-grained multi-class vehicle recognition [7], [8], [10]. Many models were trained on public datasets such as UrbanSound8K, ESC-50, and AudioSet, which contain mostly low-noise recordings from Western cities [9], [11], [13], [14]. As a result, these approaches fail to generalize in dense, noise-rich traffic environments common to South Asia. Furthermore, although Transformer-based models achieve higher accuracy, their large parameter sizes and inference latency hinder real-time deployment on edge devices [15]–[17]. To address these gaps, this paper proposes a two-tier deep learning framework for real-time horn-based vehicle detection. The first tier employs a lightweight MFCC-based CNN for horn detection. The second tier uses a Mel-

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spectrogram CNN to classify detected horns into six vehicle categories. They are: auto-rickshaw, bike, bus, car, CNG, and ambulance. This architecture reduces redundant computation. It activates the heavier classifier only upon horn detection. It ensures high accuracy with low latency on resource-limited hardware.

The rest of this paper is structured as follows. Section II explains in detail the proposed HornNet framework. Section III provides the model performance and its results. Lastly, Section IV wraps up this paper and outlines future direction.

II. Methodology

A. Data Collection

Horn sound data were collected from two densely populated regions of Dhaka. Mohammadpur and Dhanmondi selected for their high traffic density and diverse acoustic environments. These mixed residential, commercial, and institutional zones exhibit frequent horn activity. These provide varied acoustic signatures for model training. The collection sites are shown in Fig. 1.

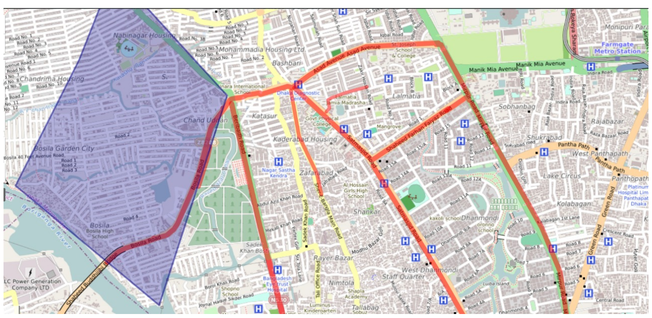

Fig. 1: Data collection zones in Dhaka: Mohammadpur and Dhanmondi. The shaded areas indicate horn recording regions.

TABLE I: Dataset Characteristics and Collection Parameters

Parameter	Specification
Collection Locations	Mohammadpur, Dhanmondi (Dhaka)
Total Recordings	2472
Vehicle Classes	6
Audio Format	WAV (16-bit PCM)
Sampling Rate	44.1 kHz
Recording Device	Android Smartphone
Average Duration	10–30 s
Environmental Conditions	Varied (peak/off-peak hours)
Distance Range	5–50 m
Background Noise	Urban traffic, crowd, market sounds

*Collected under real urban traffic conditions in Dhaka.

A total of 2,472 horn recordings were obtained across six vehicle classes—auto-rickshaw, bike, bus, car, CNG (Compressed Natural Gas three-wheeler), and *ambulance*. Dataset characteristics and recording parameters are summarized in Table I. Recordings were captured using an Android smartphone with an omnidirectional microphone positioned 5–50 m from roadways during both peak and off-peak hours to capture diverse background noise such as engines, traffic, and crowds. All audio was stored in uncompressed 16-bit PCM WAV format at 44.1 kHz with average clip lengths of 10–30 s.

B. Dataset Preprocessing and Segmentation

Each raw traffic recording was analyzed to isolate horn-centric segments suitable for model training. Fig. 2 illustrates the energy-based segmentation approach, where dynamic thresholding on short-time energy automatically identified horn-active regions within longer recordings, preserving about two-second windows around each event while excluding inactive intervals.

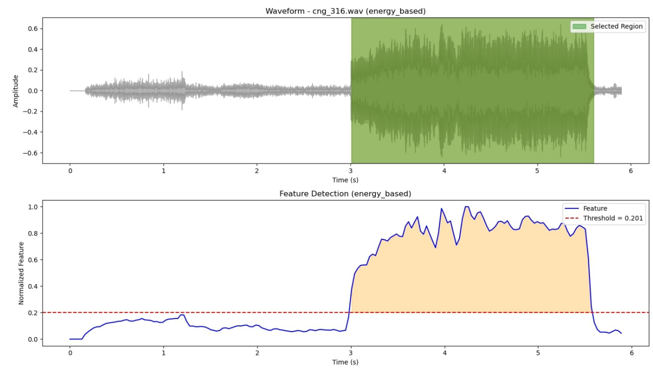

Fig. 2: Energy-based horn detection showing (a) original waveform, (b) normalized RMS energy with dynamic threshold, and (c) detected horn regions.

To enhance spectral resolution, a pre-emphasis filter $y[n] = x[n] - 0.97x[n-1]$ was applied to boost high-frequency components often attenuated in real-world recordings. All samples were then amplitude-normalized and resampled to 44.1 kHz for uniformity, as shown in Fig. 3.

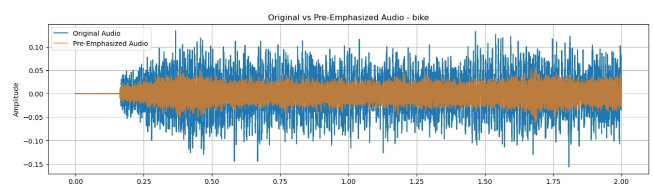

Fig. 3: Pre-emphasis comparison: (top) original signal, (bottom) pre-emphasized signal highlighting enhanced high-frequency components.

Before time–frequency transformation, several windowing functions were compared to minimize spectral leakage during Short-Time Fourier Transform. The Hamming window was selected for all subsequent processing (Fig. 4). It offers an optimal trade-off between frequency resolution and side-lobe suppression.

Each preprocessed segment was converted into two spectral representations. One is 13-dimensional MFCCs with 40 mel filters for Tier 1 horn detection. And the other 128-band log-Mel spectrograms (0–8 kHz) for Tier 2 classification. These features capture both temporal and harmonic cues whis i essential for horn-based vehicle recognition.

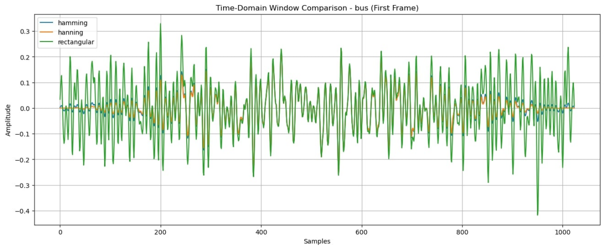

Fig. 4: Comparison of window functions: Hamming, Hanning, and Rectangular

C. Feature Extraction

Short-Time Fourier Transform (STFT) analysis represented horn energy distribution over time and frequency. It uses a Hamming window of 1024 samples with a 512-sample hop for smooth frame transitions and minimal spectral leakage. Scaled features were adopted because linear frequency scale of STFT does not match human auditory perception. The Fig. 5 represents various acoustic features.

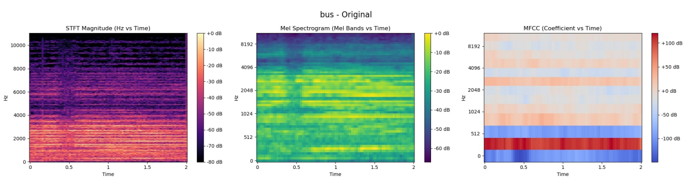

Fig. 5: Feature extraction pipeline: (left) STFT spectrogram, (middle) log-Mel spectrogram with 128 Mel bands, and (right) 13-dimensional MFCCs illustrating the transformation from detailed time–frequency to compact perceptual features.

The STFT spectra were mapped onto 128 Mel filter banks spanning 0–8 kHz to generate log-Mel spectrograms. From these, 13-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) were derived using a discrete cosine transform (DCT) on log-energy bands. For each two-second, the MFCCs for Tier-1 horn detection resulted in a 13×87 feature matrix. These features were normalized to have zero mean and unit variance to be device- and environment-independent.

D. Data Augmentation

Several augmentation techniques were applied to improve model robustness and overcome dataset limitations. It diversifies the acoustic samples while preserving their horn characteristics. Temporal augmentation was implemented through random time-stretching. It is within ±10% of the original duration and pitch-shifting by ±2 semitones. This simulates variations in horn duration and frequency across different vehicles and recording devices. Background noise was blended to reflect real-world urban conditions. The SNR of each sample is between 10–20 dB. It incorporates ambient sounds such as engines, voices, and traffic. Additionally, random temporal shifts of up to 200 ms were introduced to account for onset misalignment. The amplitude scale is between 0.8× and 1.2× was applied to model. It differences in loudness due to varying distances and microphone sensitivities. These augmentations significantly increased data diversity, reduced overfitting, and improved the model's generalization.

E. Proposed System Framework

The complete process of the proposed HornNet framework is demonstrated in Fig. 6. The system processes continuous urban audio streams to classify vehicles in real-time using horn sounds based on the two-tier deep-learning framework. The system processes the audio content using Tier-1, the lightest classifier, to determine whether there are horn sounds or not in the audio content, with the audio content being discarded, in case there are no horn sounds, to conserve system resources, while in the case of horn sounds, the system uses Tier-2 for vehicle classification studies.

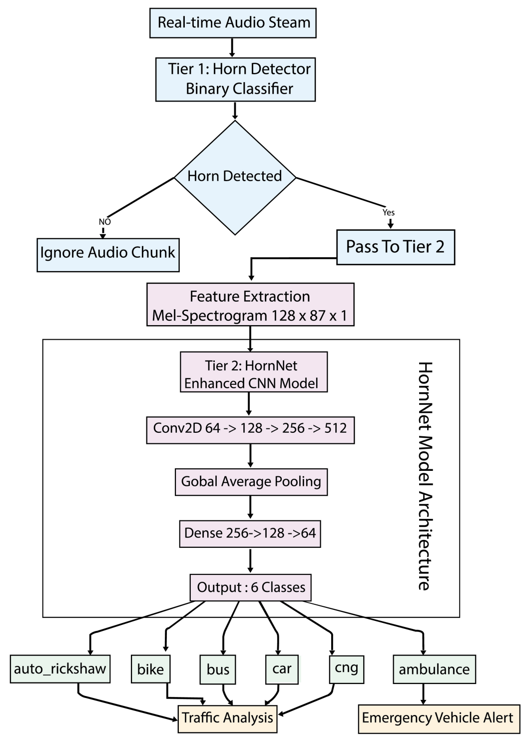

Fig. 6: The Proposed System Framework.

Also, before the classification task, Mel-spectrogram feature extraction of size $128 \times 87 \times 1$ is performed for the horn segment identified, in order to retain meaningful spectral information. Tier 2, an improved version of the convolutional neural network (CNN) approach, is employed for horn source classification across six types of vehicles: auto-rickshaw, bike, bus, car, CNG, and ambulance. The output of the classification task can further be utilized with additional modules for processing in real-world settings for real-time traffic monitoring analysis or alerting systems for emergency vehicles.

F. Two-Tier Architecture Design

HornNet uses a two-tier hierarchical structure that maximizes accuracy and efficiency in real-time horn-based vehicle classification. Tier 1 has a small CNN that performs binary horn detection based on 13-dimension MFCC features extracted from two-second clips. Once a horn event has been detected, Tier 2 kicks in and performs precise vehicle classification based on 128-band log Mel spectrogram input features. The CNN in Tier 2 has a deeper structure and features four convolutional layers (kernels 64, 128, 256, 512), accompanied by batch layers, dropout layers, and a global average layer. The fully connected layers in Tier 2 are designed with 256, 128, and 64 units, producing a softmax output that identifies six classes: auto-rickshaw, bike, bus, car, CNG, and ambulance.

G. Training Methodology

The dataset was partitioned at the recording level to prevent segment-level leakage. Entire recordings were assigned exclusively to either training (60%), validation (20%), or testing (20%) sets. Recordings from both collection regions (Mohammadpur and Dhanmondi) were proportionally distributed across splits to preserve environmental diversity. No horn segment from a single original recording appeared in multiple subsets. This recording-level separation ensures unbiased evaluation and reduces overfitting to location-specific acoustic patterns. Data augmentation techniques like time-stretching, pitch modification, noise injection, time shifting, and amplitude scaling improved the generalization capabilities. Standardization of all the features on a zero-mean and unit variance scale improved the performance. The Adam optimizer with a 1×10^{-3} learning rate and the categorical cross-entropy loss function are employed. Additionally, the models utilized early stopping on the validation accuracy. They trained the models up to a maximum of 100 epochs with a batch size of 32. The Tier-1 model filters the non-horn audio with Tier-2-activation just on the detection of the horn audio. Although data were collected within Dhaka, the inclusion of varied traffic densities, peak/off-peak conditions, and heterogeneous background noise improves environmental diversity; however, cross-city validation remains future work.

III. Results and Analysis

This section presents a comprehensive evaluation of the proposed HornNet framewor. It includes performance metrics, comparative analysis with baseline methods, ablation studies, and computational efficiency assessment.

A. Overall System Performance

The overall system performance (91.3%) in Table II denotes the end-to-end accuracy. It is consistent with the product of Tier 1 and Tier 2 accuracies (96.2% × 94.7% = 91.1%).

TABLE II: Overall System Performance of HornNet Framework

Component	***Acc.***	***Prec.***	***Rec.***	***F1***
Tier 1: Horn Detection	96.2%	95.8%	96.5%	96.1%
Tier 2: Vehicle Classification	94.7%	94.9%	94.9%	94.9%
End-to-End System	**91.3%**	**91.5%**	**91.2%**	**91.3%**

B. Class-wise Performance Analysis

Table III presents detailed performance metrics for each vehicle class, demonstrating the model's consistent performance across all categories.

TABLE III: Class-Wise Performance Analysis of HornNet

Vehicle Class	***Precision***	***Recall***	***F1-Score***	***Accuracy***
Auto Rickshaw	0.956	0.941	0.948	94.1%
Bike	0.932	0.947	0.939	94.7%
Bus	0.961	0.955	0.958	95.5%
Car	0.943	0.938	0.940	93.8%
CNG	0.935	0.949	0.942	94.9%
Ambulance	0.968	0.962	0.965	96.2%
Macro Average	**0.949**	**0.949**	**0.949**	**94.7%**

C. Confusion Matrix Analysis

The confusion matrix in Fig. 7 summarizes HornNet's class-wise prediction performance. Most samples align along the diagonal, indicating strong separability among vehicle classes and minimal inter-class confusion.

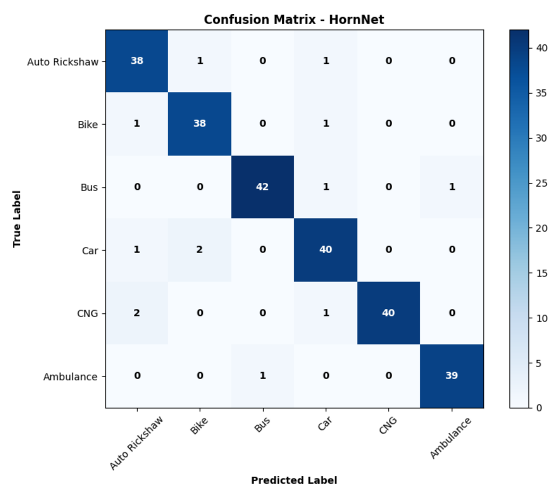

Fig. 7: Confusion matrix of the proposed HornNet framework across six vehicle categories.

HornNet achieved high true-positive rates across all six categories, with particularly strong discrimination for Emergency Vehicle and Bus classes due to their distinct horn patterns. Minor confusion occurred between Two-Wheeler and Auto-Rickshaw, mainly from overlapping frequency ranges and background noise. This is primarily attributable to overlapping spectral envelopes in the 1–3 kHz range and similar temporal horn bursts under dense traffic noise. Such overlap suggests that purely acoustic harmonic cues may be insufficient in certain edge cases, motivating future integration of contextual or multimodal features.

D. ROC-AUC Curve Analysis

To further assess HornNet's discriminative capability, Receiver Operating Characteristic (ROC) curves were generated for all six vehicle categories, as shown in Fig. 8. The corresponding Area Under the Curve (AUC) values represent the model's classification confidence across varying thresholds.

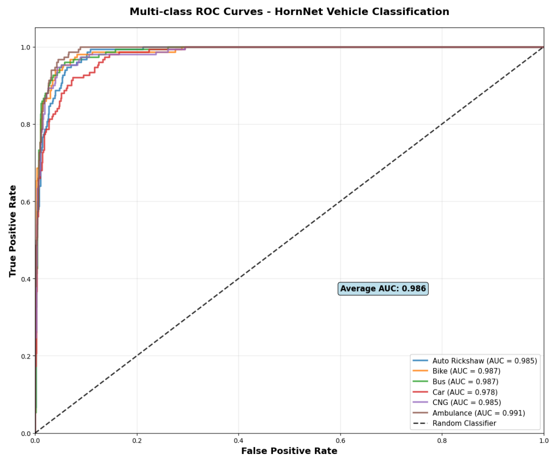

Fig. 8: ROC-AUC curves for all six vehicle categories generated by the HornNet framework.

All classes achieved AUC values above 0.96, indicating excellent separability of horn-based acoustic features. Emergency Vehicle and Bus categories showed the highest AUC scores due to distinct horn patterns, while slightly lower values for Two-Wheeler and Auto-Rickshaw resulted from overlapping frequency bands. Overall, the results confirm HornNet's strong discrimination and robustness under diverse urban noise conditions.

E. Ablation and Robustness Study

Table IV shows that by removing temporal convolution layers or spectral augmentation significantly reduced performance. It confirms their importance for accurate acoustic feature extraction.Computational load reduction was measured as the relative decrease in average inference operations per audio stream compared to a single-tier full-classification pipeline. In HornNet, Tier 2 classification is activated only upon Tier 1 horn detection. Given the observed horn occurrence rate of 21.7% in continuous recordings, this conditional activation reduced Tier 2 executions by 78.3%.

TABLE IV: Ablation Study on HornNet Components

Configuration	**Accuracy (%)**
Full HornNet (Proposed)	**94.7**
Without Temporal Convolution	91.6
Without Spectral Augmentation	92.4
Single-Tier Architecture	90.2
MFCC Features Only	89.8

To test noise resilience, HornNet was evaluated under different environmental sound conditions. The model maintained high accuracy even under adverse noise levels. It is summarized in Table V.

TABLE V: Robustness Evaluation under Noise Conditions

Noise Scenario	**Accuracy (%)**
Quiet Street (Baseline)	94.7
Heavy Traffic Noise	91.8
Construction Noise	90.6
Pedestrian and Market Noise	89.7
Low SNR (10 dB) Environment	89.0

Finally, Table VI presents the comparative evaluation against baseline methods. HornNet achieves the best trade-off between accuracy, inference time, and model size. It confirming its suitability for real-time applications.

TABLE VI: Comparative Analysis with Baseline Methods

Method	**Acc. (%)**	**F1 (%)**	**Inf. Time (ms)**	**Model Size (MB)**
SVM [10]	87.3	86.9	15.2	2.1
1D-CNN [13]	92.1	91.8	28.7	8.5
ResNet-50	93.8	93.5	45.3	94.2
Transformer-CNN [7]	94.2	93.9	52.1	127.6
HornNet (Ours)	**94.7**	**94.9**	**9.8**	**8.7**

F. Deployment Feasibility and Runtime Evaluation

HornNet was evaluated on Raspberry Pi 4, NVIDIA Jetson Nano, and Coral TPU platforms. The results in Table VII show that the model achieves efficient inference with low latency and moderate memory use. It enables real-time operation in acoustic monitoring systems. For each device, inference latency and power consumption were measured over 50 consecutive runs and reported as mean values. Observed latency variation remained within ±0.6 ms on GPU-based platforms and ±1.4 ms on CPU-based platforms. Power measurements fluctuated within ±0.3 W across runs, indicating stable runtime characteristics.

TABLE VII: Deployment Performance on Embedded Platforms

Platform	**Avg. Inf. Time (ms)**	**Memory (MB)**	**Power (W)**
Raspberry Pi 4 (CPU)	42.6	97.3	4.2
NVIDIA Jetson Nano (GPU)	18.9	112.5	7.8
Coral TPU (Edge Accelerator)	**9.4**	86.2	**3.9**
Laptop i7 (Reference)	8.1	256.8	15.2

The Coral TPU offered the best trade-off. It achieves near real-time inference ($\approx$9 ms per sample) with power consumption below 4 W. Even on the Raspberry Pi 4, HornNet sustained over 20 frames per second. This demonstrates compatibility with low-power devices. These results confirm that HornNet is lightweight and energy-efficient. And it is suitable for large-scale urban traffic monitoring and smart transportation applications. While embedded deployment demonstrates real-time feasibility on representative edge platforms, large-scale scalability in distributed urban sensing networks remains to be evaluated. Future work will investigate distributed inference architectures, model quantization at scale, and cloud–edge hybrid scheduling to support city-wide deployments.

G. Discussion

Acoustic vehicle recognition presents a number of challenges in urban environments related to overlapping noise sources, reverberations, and variable horn characteristics. The proposed framework HornNet was developed to solve these problems with a two-tier deep learning design separately handling the tasks of horn detection and classification. Table II shows that HornNet's performance surpassed existing techniques such as SVM [10], 1D-CNN [13], and Transformer-CNN [7]. The hierarchical nature of the HornNet architecture allowed for better isolating horn events from the general background noise which delivers an end-to-end accuracy of up to 91.3%. The confusion matrix of Fig. 7 notices how minor confusion mainly lies along the classes of Two-Wheeler and Auto-Rickshaw. Fig. 8 exhibits an average AUC value of 0.986 that confirms the fact that HornNet preserves high confidence in multi-class classification under varying noise conditions.

Ablation analysis summarized in Table IV confirms that each component of the model contributes meaningfully to overall performance. Temporal convolution or spectral augmentation removal caused a noticeable drop in accuracy. Furthermore, it follows from robustness evaluation provided in Table V that even under challenging conditions-like heavy traffic or construction areas. HornNet demonstrates the accuracy above 89% which outperforms earlier frameworks reported in [7], [13]. Table VI shows that HornNet outperformed existing models by both accuracy and inference time, what witnesses about favorable balance between precision and computational efficiency.

In terms of deployment, Table VII shows HornNet's efficacy for embedded and real-time systems. HornNet made possible inference times of 9.4 ms on Coral TPU with very low power consumption. Nevertheless, some issues arise in horn patterns with similar acoustic characteristics, such as those of lighter vehicles and motorcycles. Future work should be directed along lines such as incorporating attention-enabled modules. It will enhance data diversity using different horn patterns, and multi-modal fusion using videos and vibrations. The aim is to improve classification accuracy and scalability for megacity environments.

IV. Conclusion

HornNet, a two-tier deep learning architecture, was proposed for real-time vehicle classification based on the acoustic signatures of horns. HornNet can detect light horns effectively and classify vehicles accurately. Experimentation showed HornNet to be robust with a detection rate of 96.2%, classification accuracy of 94.7%, and overall performance of 91.3%. For computational efficacy, HornNet performed better than the SVM model, the 1D-CNN model, ResNet-50, and the Transformer-CNN model. HornNet ran with a time complexity of 9.4 ms in Raspberry Pi and Coral TPU, consuming very low power, supporting real-time ITS deployment. Despite strong performance, challenges remain in distinguishing acoustically similar horn signatures, particularly among light vehicles. Future work will incorporate attention-based modules to better capture fine-grained temporal harmonics and explore multimodal fusion of audio with video or vibration signals. Additionally, large-scale distributed deployment and cross-city validation will be investigated to enhance robustness and scalability for megacity traffic environments.

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